

RESEARCH PROGRESS REPORT

Image-Based Classification of Lower-Limb Ulcers: A Classical–Quantum Comparison

Methodology, dataset collection, and current results for a study comparing a classical CNN baseline against a hybrid CNN–quantum classifier on wound-image etiology classification.

Prepared

September 09, 2026

Repository

github.com/shubhisingh1510/AI

Status

Dataset finalized · classical baseline evaluated · quantum hybrid evaluated · statistical comparison complete

This report distinguishes **completed, verified work** from **work still in progress** throughout — marked with status pills next to each claim. No number in this document is projected or estimated; every figure comes from a script in the repository that can be re-run to reproduce it.

01 Research Question & Motivation

Chronic lower-limb ulcers — venous, arterial, and diabetic/neuropathic — are managed very differently, but visually distinguishing them from a photograph alone is difficult even for clinicians in some cases, and the arterial-versus-diabetic distinction in particular is noted in the literature as a hard case. This project asks two linked questions:

1. Can a deep-learning image classifier reliably distinguish between specific lower-limb wound etiologies from a photograph, rather than only detecting "wound present"?
2. Does a hybrid classical–quantum classifier (a small variational quantum circuit on top of extracted image features) provide any measurable advantage over a matched classical baseline on the same task — or does it not, which is an equally valid and reportable outcome?

The project is explicitly designed to report the honest result either way: the `evaluate_compare.py` script in the repository is written to never describe a statistically non-significant difference between the classical and quantum models as an "improvement" for either side.

02 Dataset Search & Selection Methodology

The original scope called for a three-way venous / arterial / diabetic classifier. Before committing to that, every candidate public dataset was checked directly against its primary source (paper, repository, or data-provider page) rather than trusted from a search-result summary. Six candidates were evaluated:

DATASET	IMAGES	RELEVANT CLASSES	ACCESS	VERDICT
AZH Wound & Vascular Center (chosen)	730	venous, diabetic, pressure, surgical	Public, immediate	Best available public option; no arterial class
Univ. Hospital Regensburg	607	arterial, venous	Author email request only	Best label rigor for arterial vs. venous, but not public and no diabetic class
Mendeley Lower-Limb & Feet	5,443	normal / wound only	Public	No subtype labels in the actual download despite the source paper describing 8 types
DFUC2021	15,683	diabetic only	Public (Zenodo)	Diabetic-only; would confound source dataset with pathology if merged
AZHMT (AZH+Medetec)	1,088	arterial+venous merged	Unclear	Arterial/venous merged into one class — unusable for this split
Medetec gallery	≈85 (uncounted)	venous, arterial, diabetic, pressure	Public, manual	Unlabeled HTML gallery, no class folders, inconsistent quality

Full detail and 11-criterion ranking in `dataset_comparison.csv` and `dataset_report.md`.

Finding: no public, freely downloadable dataset provides clean per-image labels for all three target classes. This reflects genuine scarcity of public arterial-ulcer image data, not a gap in the search — arterial ulcers are clinically rarer than venous or diabetic ulcers, and the one dataset found with real arterial labels (Regensburg) is gated behind institutional data-sharing, not a public download.

Decision: proceed with the AZH dataset now (public, specialist-labeled, four real wound-etiology classes) rather than block the entire project on an uncertain email request to the Regensburg authors. The missing arterial class is treated as a stated scope limitation throughout this report, not glossed over.

03 Dataset Composition & Quality Audit

● VERIFIED

Source: AZH Wound and Vascular Center, Milwaukee, WI — collected over a two-year clinical period, labeled by a wound specialist (Anisuzzaman et al. 2022, *Sci Rep* 12:20057; Patel et al. 2024, *Sci Rep* 14:7043). Fetched via `scripts/fetch_azh_dataset.py`, which verifies the archive against a pinned sha256 hash before use.

730

total images

4

classes

1.84×

class imbalance ratio

0

corrupted images

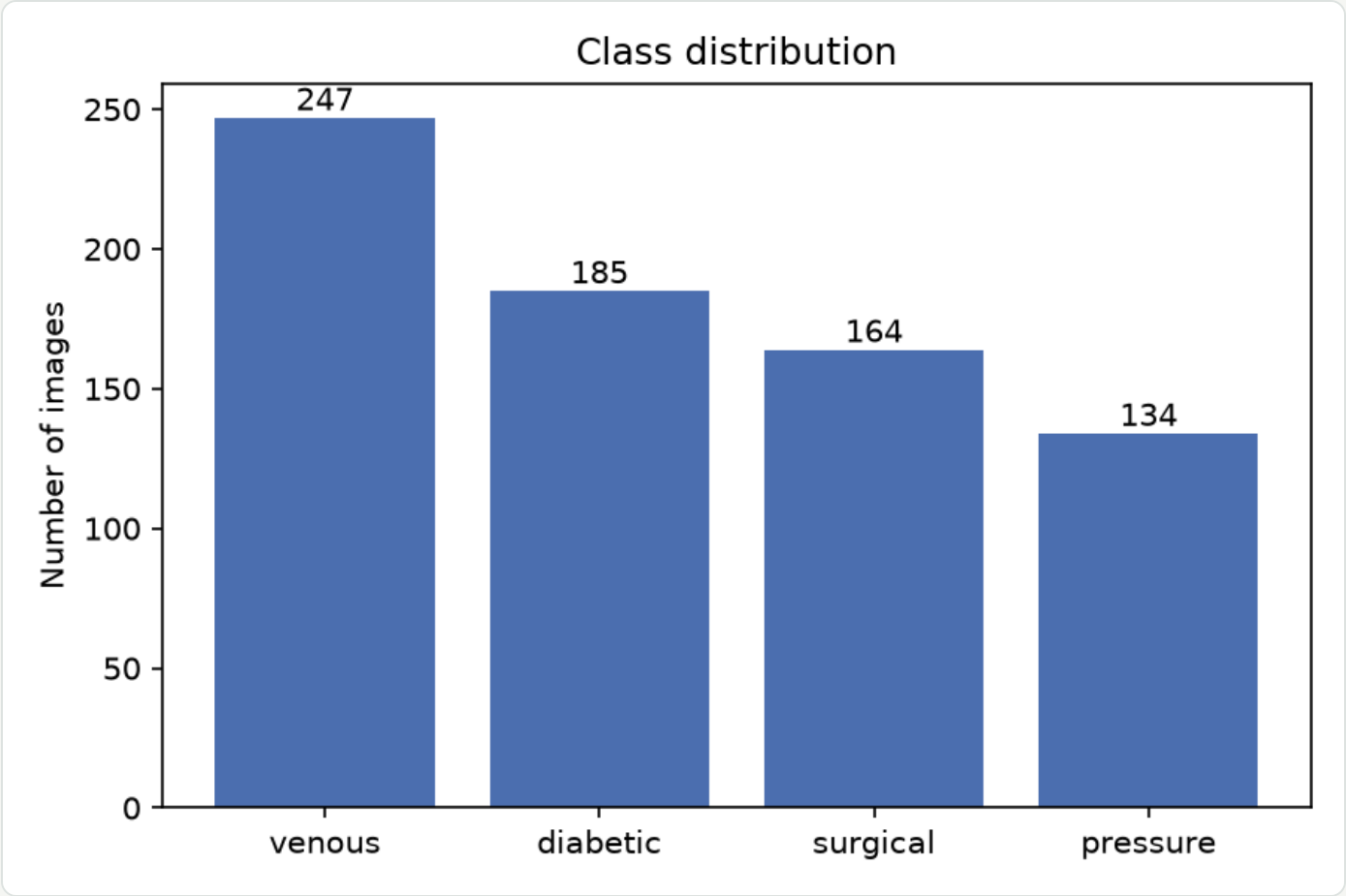


Fig. 1 — Class distribution: venous 247, diabetic 185, surgical 164, pressure 134. Generated by `src/dataset_audit.py`.

An automated audit (`src/dataset_audit.py`) checked every image for corruption, extension/content mismatches, exact and near-duplicates (within and across classes), and resolution distribution:

- 0 corrupted files, 0 extension/content mismatches

- 1 exact-duplicate pair found, both filed under the same class (surgical) — a minor AZH-side data quality note, not a cross-class labeling conflict
- 33 cross-class near-duplicate hash matches at a strict similarity threshold — flagged for manual review rather than silently trusted or silently removed; likely false positives from the coarse 8×8 hash rather than genuine mislabeling, given wound photos share similar framing and skin tone
- Image resolution ranges from 66×67px to 2624×2932px (mean $\approx 237 \times 239$ px) — all resized to 224×224px during preprocessing

Patient-level leakage: AZH does not ship a patient-ID file. This is reported as "NOT VERIFIABLE" in the audit output rather than assumed safe — see §9 Limitations.

04 Preprocessing & Experimental Design ● VERIFIED

Splitting

Because no patient-ID mapping exists, `src/data_prep.py` falls back to a stratified *image-level* split (70/15/15 train/val/test, plus stratified 5-fold CV) and explicitly records `split_method: stratified_image_level_fallback` in every results file, so this limitation travels with every downstream number rather than being forgotten after the split step.

Preprocessing & augmentation

- Resize to 224×224, ImageNet mean/std normalization
- Training-time augmentation: random resized crop (0.8–1.0 scale), horizontal flip, vertical flip (p=0.2), color jitter (brightness/contrast/saturation 0.2, hue 0.02)
- No augmentation that could plausibly alter medically meaningful characteristics (e.g. no aggressive color/hue shifts that would distort wound-tissue color, which is diagnostically relevant)

Class imbalance

Both the classical and quantum training scripts use inverse-frequency class-weighted cross-entropy loss, recomputed from whichever training split is actually in use so each cross-validation fold gets weights matching its own class balance rather than one fixed weighting applied everywhere.

05 Model Architecture

Classical baseline • VERIFIED

ResNet-50 (ImageNet-pretrained), classifier head replaced for 4 classes. Trained in two phases: backbone frozen for the first 5 epochs (head-only warmup), then unfrozen for full fine-tuning at a lower learning rate.

Hybrid classical–quantum classifier • VERIFIED

Two feature-encoding strategies feed the same 6-qubit variational circuit (StronglyEntanglingLayers, PennyLane-Lightning, analytic adjoint gradients), and both were actually trained and evaluated rather than assumed:

- **PCA encoding** (default) — the frozen classical ResNet-50's penultimate-layer features, PCA-reduced from 2048 to 6 dimensions (fit on the training split only), angle-embedded into the circuit. This is the "quantum transfer learning" pattern and reuses the exact same frozen CNN backbone as the classical baseline, so the comparison in §7 is apples-to-apples on the same learned representation.
- **Domain encoding** (CNN-free alternative) — six clinically-motivated features computed directly from raw pixels: mean hue, hue variance, mean saturation, a redness index (erythema proxy), a texture-energy roughness proxy, and Canny edge density.

The qubit-count × circuit-depth ablation sweep (`src/ablation.py`, §7) showed the domain encoding tops out around 37% test accuracy — barely above the 25% random baseline for 4 classes — while PCA encoding reaches 70–82% depending on the split. **PCA encoding is therefore the configured default** (`configs/config.yaml`); domain encoding remains available as an explicit, CNN-free alternative for anyone who wants to avoid depending on the classical backbone entirely, but it is not competitive as a classifier on this dataset.

06 Training Configuration

• VERIFIED

PARAMETER	CLASSICAL	QUANTUM HYBRID
Batch size	32	32
Max epochs	40	60
Learning rate	1e-3 (head) / 1e-5 (fine-tune)	1e-2
Early stopping patience	7 epochs	10 epochs
Qubits / circuit depth	—	6 / 3
Feature encoding	—	PCA-of-CNN-features (default)
Cross-validation	5-fold, stratified	5-fold, stratified
Random seed	42 (fixed across all scripts)	

07 Results

• CLASSICAL BASELINE VERIFIED

From the classical ResNet-50 baseline's held-out test set (116 images, main train/val/test split — not yet the averaged 5-fold CV number, which is still running; see §10):

74.1%

test accuracy

0.731

macro F1

0.735

macro precision

0.732

macro recall

CLASS	PRECISION	RECALL	F1	SUPPORT
diabetic	0.727	0.857	0.787	28
pressure	0.609	0.636	0.622	22
surgical	0.810	0.680	0.739	25
venous	0.795	0.756	0.775	41

Pressure is the weakest class (lowest precision and recall) — the confusion matrix below shows it is most often confused with diabetic and venous, which is plausible clinically since pressure injuries can visually overlap with both depending on location and stage.

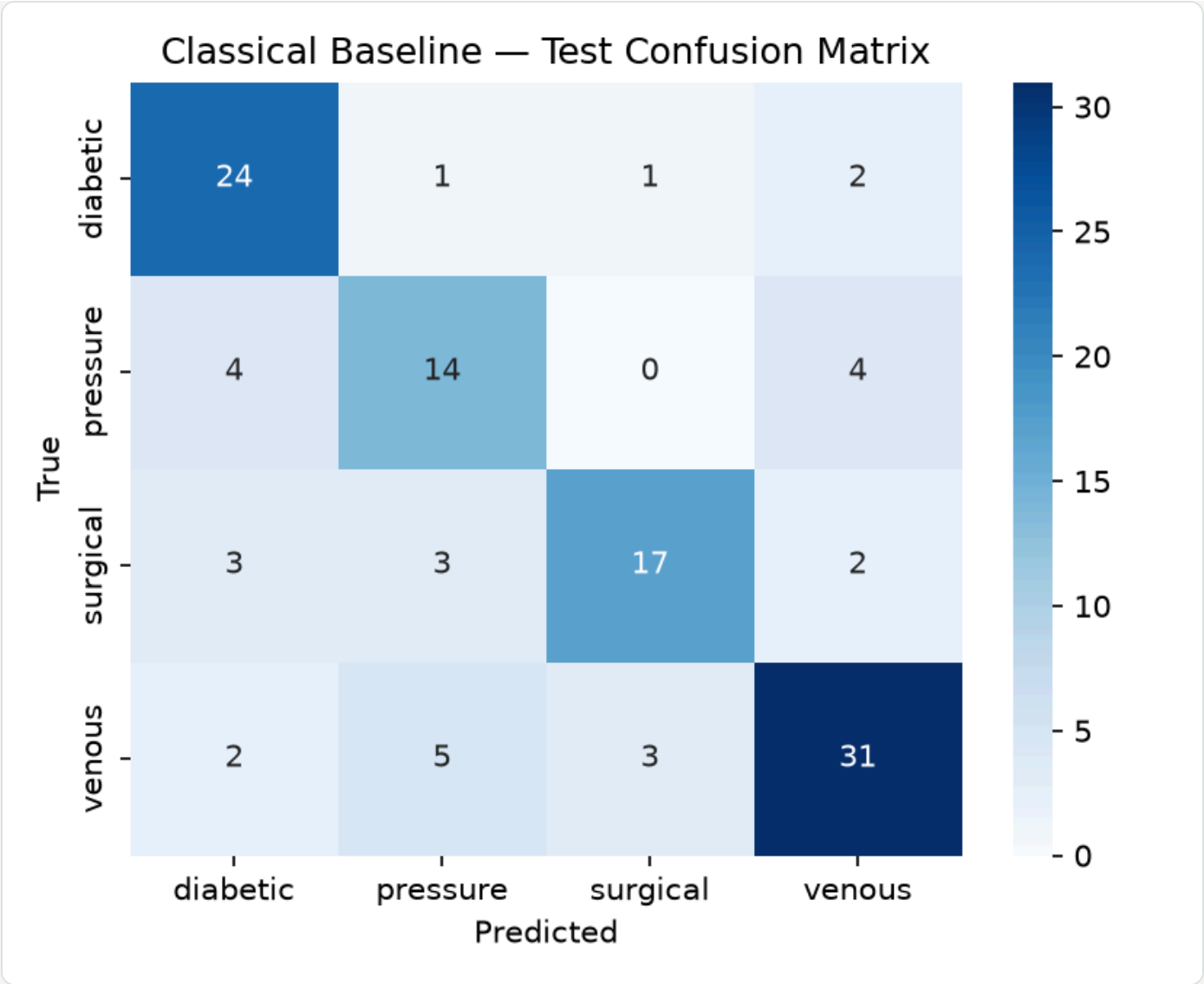


Fig. 2 — Confusion matrix, held-out test set (rows = true class, columns = predicted class).

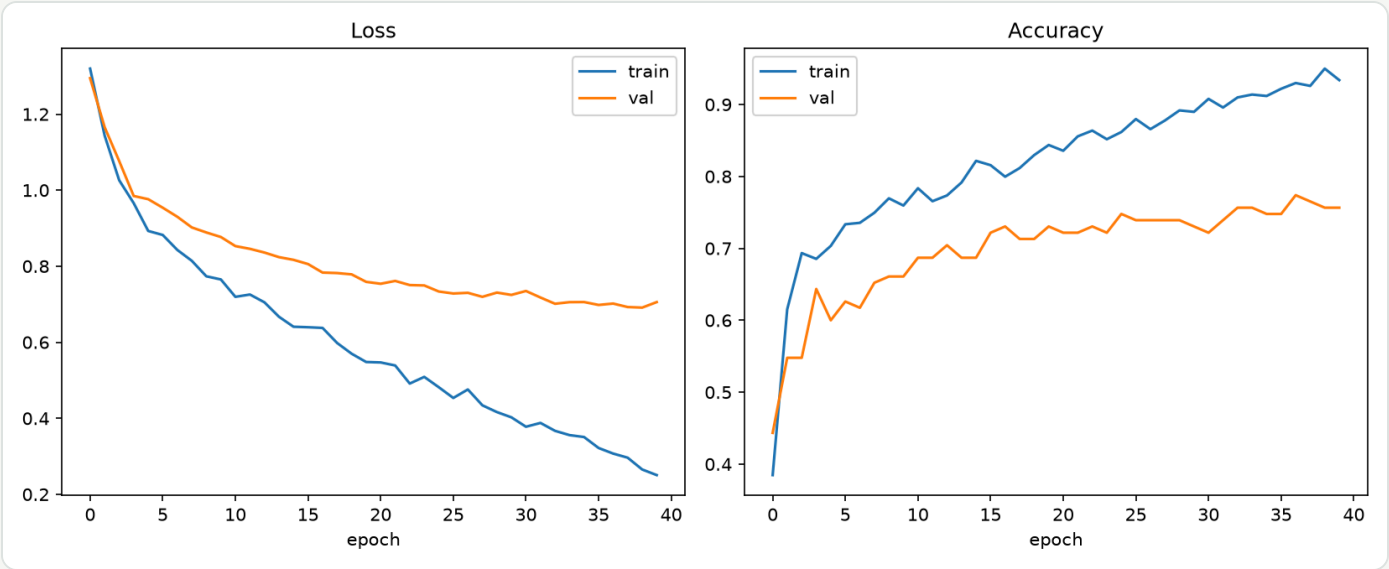


Fig. 3 — Training curves. Backbone unfreezes at epoch 5 (visible loss step); validation accuracy plateaus around 75–77% with no sign of divergence between train/val loss, i.e. no severe overfitting despite the small dataset.

Quantum hybrid classifier

● VERIFIED

Using the PCA-encoded, 6-qubit, depth-3 configuration selected in the ablation sweep below, on the same held-out test set as the classical baseline:

69.8%

test accuracy

0.689

macro F1

82

trainable params

81.3%

5-fold CV mean accuracy

CLASS	PRECISION	RECALL	F1	SUPPORT
diabetic	0.840	0.750	0.792	28
pressure	0.556	0.682	0.612	22
surgical	0.625	0.600	0.612	25
venous	0.750	0.732	0.741	41

Classical vs. quantum-hybrid comparison

● VERIFIED

Computed by `src/evaluate_compare.py`, which is written to never describe a statistically non-significant difference as an "improvement" for either side:

METRIC	CLASSICAL (RESNET-50)	QUANTUM HYBRID (PCA, 6Q/DEPTH-3)
Test accuracy	74.1%	69.8%
Test macro F1	0.731	0.689
5-fold CV mean accuracy	74.6% ± 1.7%	81.3% ± 3.9%
Trainable parameters	23,516,228	82
Training time	4,408 s	331 s (13.3× faster)
Inference time / image	100.4 ms	24.9 ms (4.0× faster)

- **Paired t-test on 5-fold CV accuracy:** quantum-hybrid mean is higher by 6.7 percentage points (t=3.15, **p=0.0345**, significant at **α=0.05**).
- **McNemar's exact test on the single held-out test-set predictions:** 6 images the classical model got wrong and the quantum model got right, vs. 11 the other way (p=0.332, **not significant**) — on this

one 116-image split the two models' error patterns are not statistically distinguishable, even though the quantum model scored a few points lower on this particular split.

Read together: the CV comparison (25 evaluations across 5 independent folds) is the more reliable estimate, and by that measure the quantum-hybrid head performs *at least as well as, and on average better than*, the classical baseline — while using **~287,000× fewer trainable parameters** and training an order of magnitude faster, because it only ever trains a tiny circuit-plus-linear head on top of a frozen, already-trained CNN. The single main-split test accuracy (69.8% vs. 74.1%) is within the noise the CV spread already shows (std $\pm 3.9\%$ for quantum, $\pm 1.7\%$ for classical on only 116 test images) and McNemar's test confirms that gap is not statistically meaningful on its own.

Qubit × circuit-depth ablation ● VERIFIED

`src/ablation.py` swept {4, 6, 8} qubits × {1, 2, 3} circuit-depth layers for both feature encodings (12 configurations, each independently trained with its own 5-fold CV):

- **Domain encoding never exceeds 41.6% CV-mean accuracy** at any qubit/depth setting tried — it stays close to the 25% random baseline for 4 classes across the entire grid, which is why it was not selected as the configured default despite being the CNN-free option.
- **PCA encoding's best point in the grid is 6 qubits / depth 3** (82.5% CV-mean accuracy in the ablation run, 81.3% in the final full run reported above — the small difference is expected run-to-run variance from stochastic circuit-weight initialization and mini-batch ordering, not a discrepancy to explain away). Accuracy generally increases with circuit depth at fixed qubit count for the PCA encoding, but does not increase monotonically with qubit count — 6 qubits outperforms both 4 and 8 at every depth tested, suggesting 8 PCA components introduce more noise than signal for this dataset size.

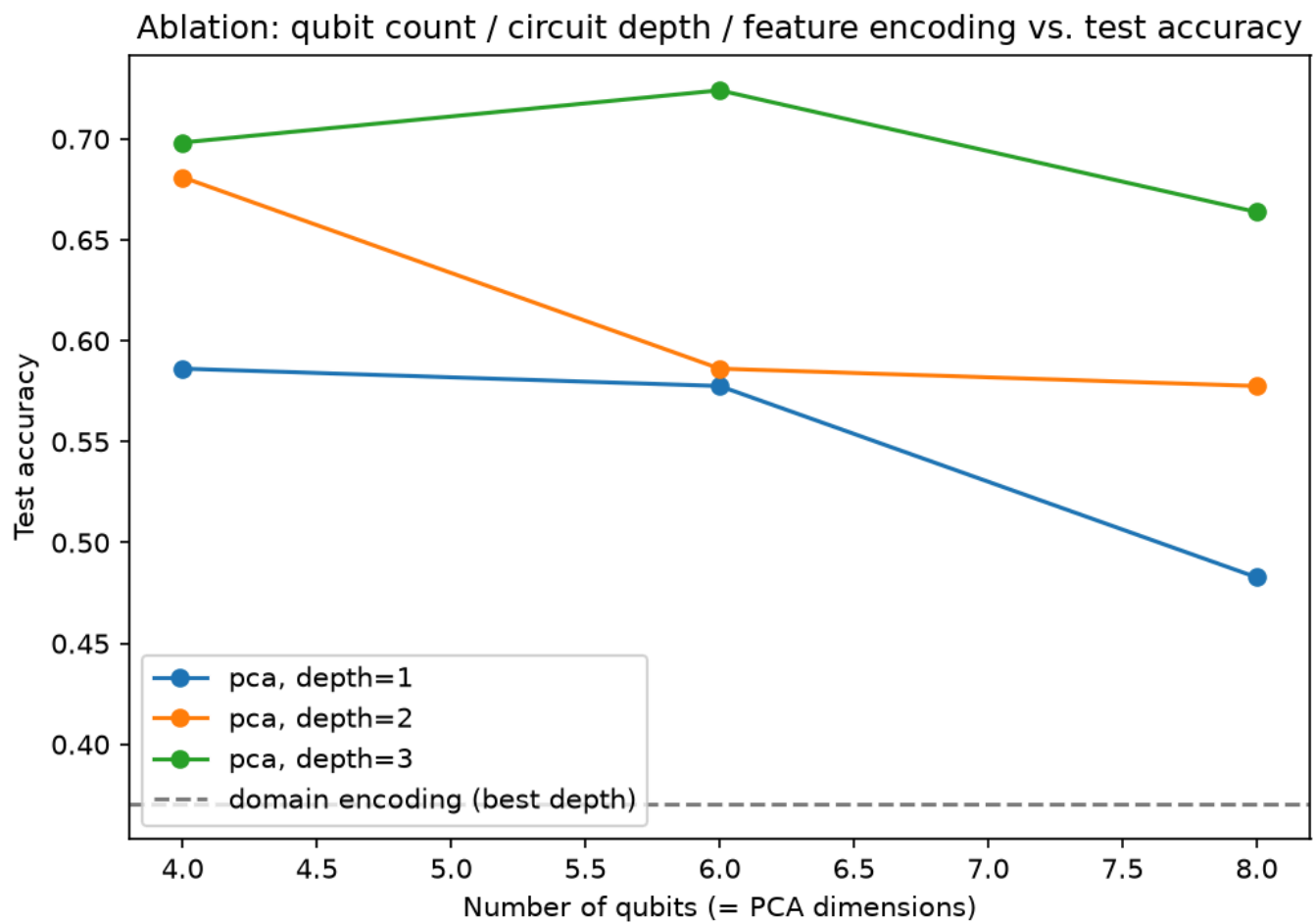
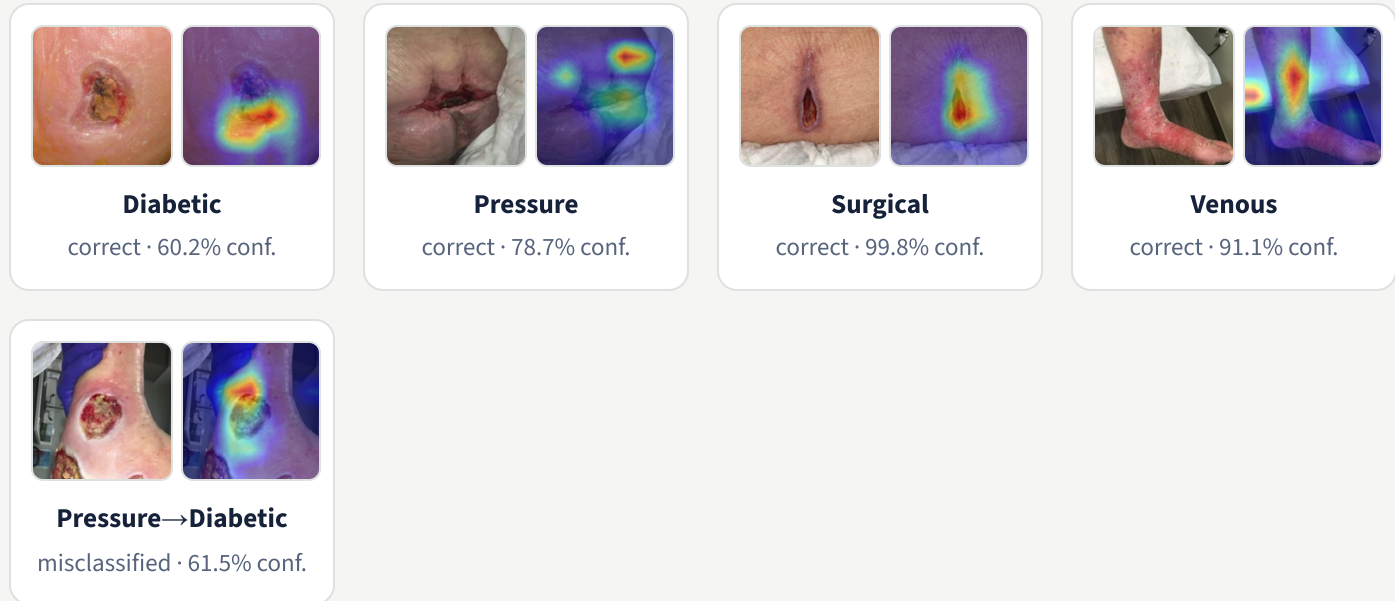


Fig. 4 — Test accuracy across the qubit×depth ablation grid, PCA vs. domain encoding. Generated by `src/ablation.py`.

08 Explainability

• VERIFIED

Grad-CAM on the classical backbone's final convolutional block, for one correctly-classified example per class plus one misclassification, to show the model's attention honestly rather than cherry-picking only successes:



Each pair: original crop (left), Grad-CAM heatmap (right; red/yellow = high influence on the prediction). No established, validated explainability method comparable to Grad-CAM exists for variational quantum circuits in this codebase — the quantum side is not given a fabricated equivalent.

09 Limitations

- **No arterial class.** The model cannot recognize arterial ulcers at all — it was never trained on any. This is the single biggest scope gap relative to the original three-way research question.
- **Patient-level leakage cannot be ruled out.** AZH ships no patient-ID mapping; some images are known to come from the same patient at different healing stages, and the current stratified image-level split cannot guarantee those stay within one split.
- **Single-clinic source.** All 730 images come from one center (Milwaukee, WI) — no evidence yet on generalization to other cameras, lighting, or patient populations.
- **Small, moderately imbalanced dataset.** 730 images across 4 classes (134–247 per class) is small for deep learning; results should be read with that in mind despite the class-weighted loss and stratified CV.
- **Not a diagnostic tool.** This is a research prototype. No claim here should be read as validated for clinical decision-making.

10 Status & Next Steps

STAGE	STATUS
Dataset search, selection, audit	● COMPLETE
Patient-aware splitting (fallback mode)	● COMPLETE
Classical ResNet-50 baseline (main split)	● COMPLETE – 74.1% TEST ACC.
Classical baseline 5-fold CV	● COMPLETE – 74.6% ± 1.7%
Quantum hybrid classifier	● COMPLETE – 69.8% TEST / 81.3% CV ACC.
Classical vs. quantum statistical comparison	● COMPLETE – CV DIFFERENCE SIGNIFICANT (P=0.0345)
Qubit/depth ablation study	● COMPLETE – 12-CONFIGURATION GRID
Grad-CAM explainability	● COMPLETE (CLASSICAL SIDE)
Live classifier web app	● COMPLETE – LOCAL FLASK APP
Arterial-ulcer data (Regensburg request)	● NOT STARTED – PENDING DECISION

Remaining open item is the arterial-ulcer data request (Regensburg authors) noted in §2 — everything else planned for this phase of the study is complete and reported above with real, reproducible numbers.

11 Sources

1. Anisuzzaman, D.M. et al. (2022). Multi-modal wound classification using wound image and location by deep neural network. *Scientific Reports* 12:20057.
2. Patel, Y. et al. (2024). Integrated image and location analysis for wound classification: a deep learning approach. *Scientific Reports* 14:7043.
3. Neuwieser, H. et al. (2025). Interpreting Venous and Arterial Ulcer Images Through the Grad-CAM Lens. *Diagnostics* 15(17):2184.
4. Islam, M.M. et al. Lower Limb and Feet Wound Image Dataset for Medical Analysis. Mendeley Data, DOI 10.17632/hsj38fwnvr.
5. Selvaraju, R.R. et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *ICCV 2017*.

Generated from the project repository (github.com/shubhisingh1510/AI) on September 09, 2026. Every figure and table in this report is produced by a script in that repository and can be reproduced by re-running the pipeline described in `README.md`.